DAY 21 | POST-READ

GenAI and Prompting: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
LLM loop	Tokenise, embed, transform with attention, predict the next token, repeat.
Tokens and context	Models read and write tokens; the context window limits how much fits.
Temperature	Low for factual and consistent output, higher for variety.
Attention	softmax(QK^T / sqrt(d)) V: each token weighs the others.
Architectures	Encoder-only (understand), decoder-only (generate), encoder-decoder (transform).
MoE	A router sends each token to a few experts: big capacity, lower cost per token.
Prompt anatomy	Role, task, context, format, examples, constraints.
Few-shot	Examples teach format and edge cases.
CoT / ReAct / ToT	Steps before answers; thought-action-observation with tools; explore and evaluate several paths.

Part B: Going beyond today

1. Structured outputs

Many APIs can force the model to return JSON matching a schema. Combine this with Pydantic (Day 6) to validate every response before your code uses it.

2. System prompts and messages

Chat APIs take a list of messages: system (rules and role), user (requests) and assistant (previous replies). Put stable rules in the system message and task details in the user message.

3. Self-consistency

Ask the same reasoning question several times at moderate temperature and take the most common answer. It often beats a single attempt on tricky problems, at extra cost.

4. Reasoning models

Some newer models think internally before answering. For these, give a clear goal and constraints rather than step-by-step instructions, and check their output as carefully as any other.

5. Prompts are code

Keep prompts in version control, test them against a fixed set of examples, and review changes like code. You will build prompt regression tests on Day 30.

Part C: Check yourself

- Why does an LLM sometimes give different answers to the same question?
- In your own words, what does attention do?
- When would you choose few-shot over zero-shot?
- What are the three steps that repeat in ReAct?
- Why can MoE models be large yet efficient?

Part D: Practice ideas for tonight

- Rewrite a vague prompt from your work using all six prompt parts and compare outputs.
- Add a third tool (unit converter) to the ReAct agent in the notebook.
- Use Tree of Thought to plan a three-day study schedule for Week 6.

Coming up on Day 22: context engineering and RAG: what to put into the model's context, system architecture for AI apps, and chunking documents.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today